

The Novel Profiling Relative Inhibition Simultaneously in Mixtures (PRISM) Platform Identifies Synergistic Activity of Lanraplenib and Ruxolitinib in Hematological Malignancies

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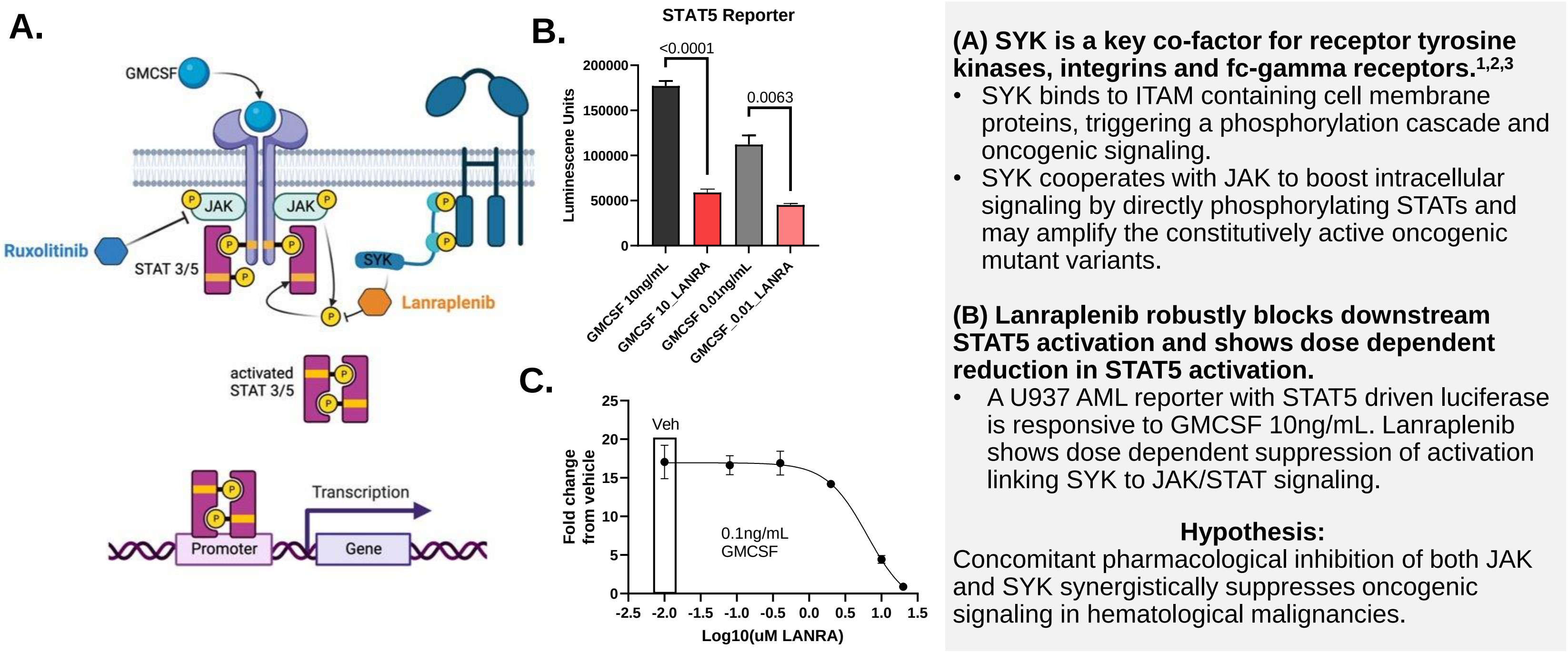
Abstract

Background: Investigation of drug combinations across different contexts can provide useful insights on the anti-cancer mechanism and can ultimately lead to new treatments. However, conventional drug combination screening methods are limited by throughput. Efforts to systematically identify the most effective active combinations and the optimal molecular contexts in a high throughput screening (HTS) format could greatly accelerate the development of combination treatments. Spleen Tyrosine Kinase, SYK, is a non-receptor tyrosine kinase known to regulate intracellular signaling, including FLT3, AKT/mTOR and STAT5 pathways, via its immunoreceptor tyrosine-based activation motif (ITAM). Deregulated SYK signaling has been reported to play a central role in pathogenesis of allergic and autoimmune diseases as well as hematological malignancies. Lanraplenib (LANRA) is a next-generation SYK inhibitor currently being evaluated in combination with gilteritinib, a FLT3 inhibitor, in patients with relapsed or refractory (R/R) *FLT3*-mutated acute myeloid leukemia (AML) (NCT05028751). Given its critical role in intracellular signaling and interaction with receptor tyrosine kinases (RTKs), we hypothesized that LANRA could synergize with ruxolitinib, a JAK inhibitor. To address this hypothesis, we performed a high throughput drug combination screen using the Broad Institute's Profiling Relative Inhibition Simultaneously in Mixtures (PRISM) platform, which enables rapid screening of thousands of compounds in a 930-cell line panel across 45 different lineages.

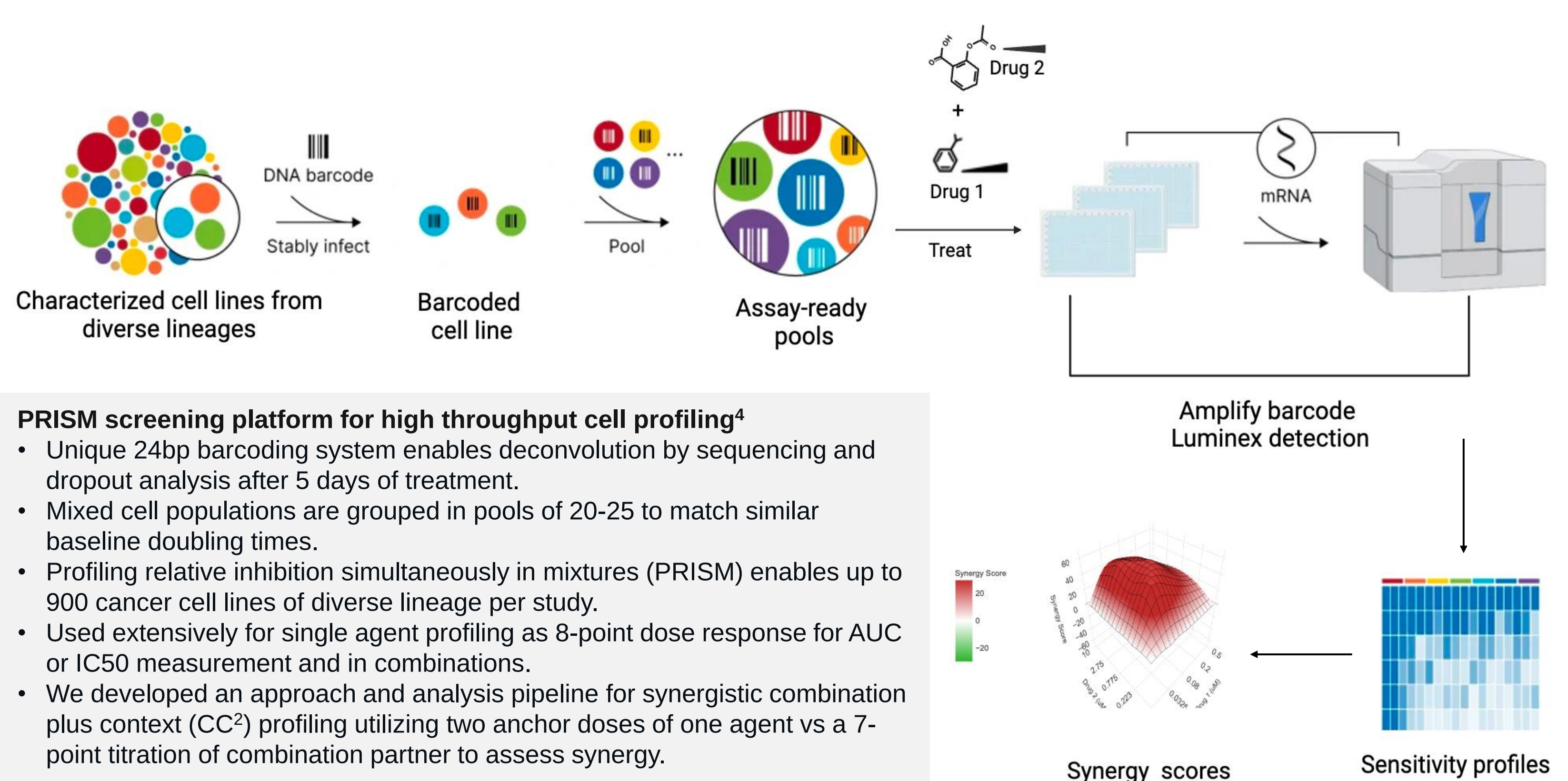
Methods: In this study, we performed a combination screen using the PRISM platform with ruxolitinib as the test compound, with a 10 μ M top dose (7 dose concentrations with 3-fold dilutions), in combination with LANRA at two anchor doses, 2 μ M and 10 μ M. PRISM is a pooled, multiplexed cell viability assay that provides 7-point dose response curves, IC, AUC values, and relative abundance of unique cell line barcodes. To understand the drug synergy landscape across different lineages, we developed a bioinformatics pipeline which uses PRISM viability data to calculate synergy scores across all the cell lines and drug combinations. Secondary validation studies of the combinations used CTG viability measurements. Phospho-SYK expression was evaluated in archival formalin-fixed paraffin-embedded (FFPE) bone marrow biopsies from patients with myeloid proliferative neoplasm (MPN) by immunohistochemistry (IHC). RNA-seq was performed to evaluate differential changes in gene expression in response to LANRA. Gene set enrichment analysis (GSEA) was performed to evaluate perturbation in leukemogenic signaling pathways.

Results: In the PRISM cell line panel, ruxolitinib in combination with LANRA demonstrated synergistic activity in hematological malignancy cell lines. Among AML cell lines in the panel, OCIAML5, OCIM1, HL60, HEL, EOL1, MONOMAC6, NB4, U937, PL21, and TF1 showed the highest synergy scores. The most sensitive cell lines to the combination showed up-regulation of JAK-STAT and inflammatory signaling pathways in a gene set enrichment analysis (GSEA) prior to treatment. Consistent with this, phosphorylated SYK was associated with inflammatory megakaryocytes and fibrosis in primary samples from patients with MPN. LANRA down-regulated JAK-STAT signaling in a reporter cell line and repressed gene expression associated with dysregulated inflammatory pathways in AML cells. Additionally, the combination of LANRA and ruxolitinib showed synergistic antiproliferative activity across a broad range of concentrations in FLT3-ITD primary AML cells and a panel of hematological malignancy cell lines, confirming the PRISM results.

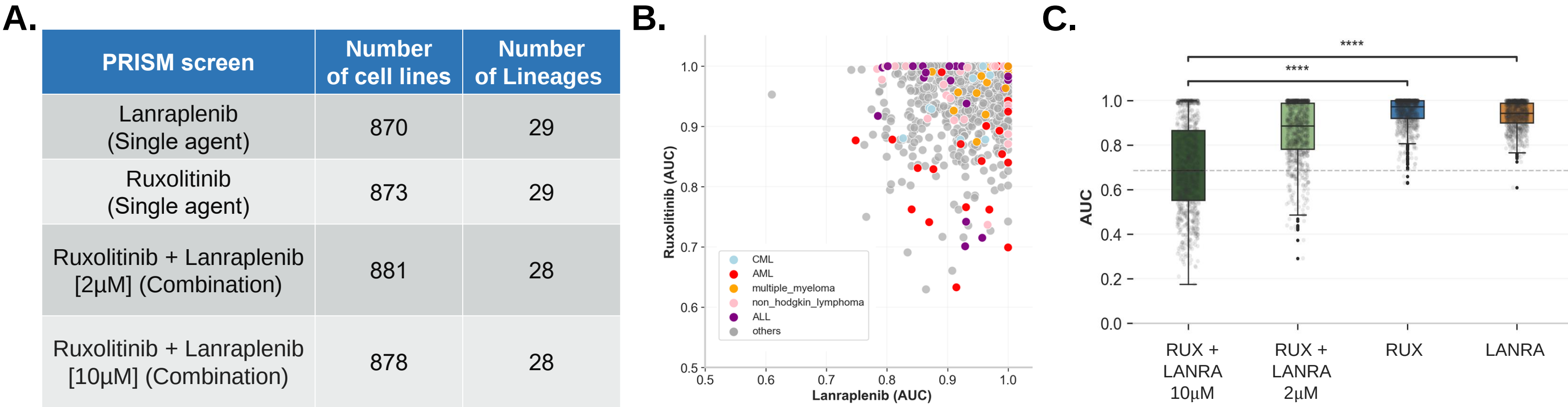
SYK and JAK/STAT signaling cooperate to drive oncogenic signaling



PRISM enables the generation of high throughput anti-proliferative profiling



PRISM combination screen identifies synergy in Ruxolitinib and Lanraplenib combination



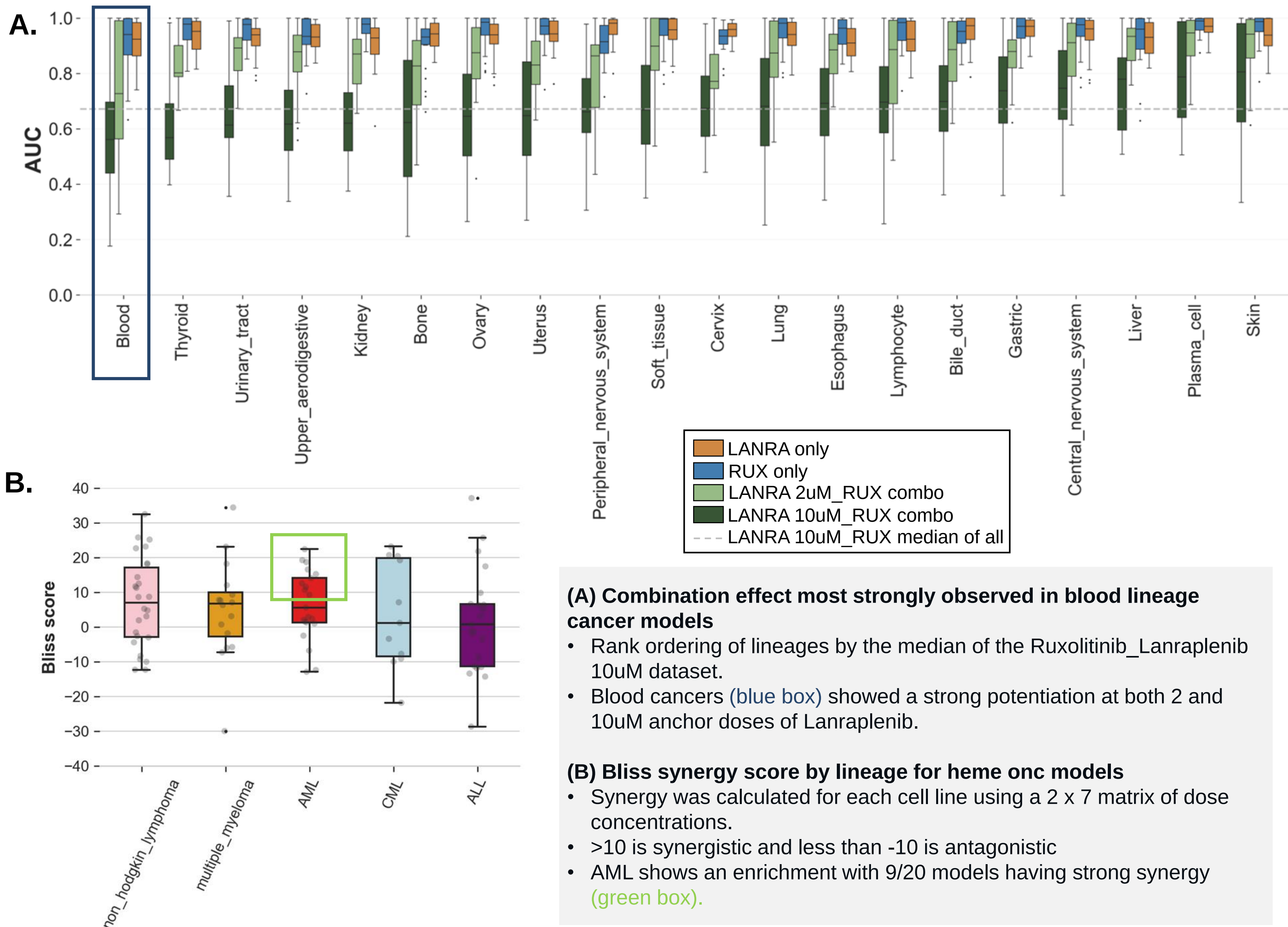
Single agent profiling identified heme models as sensitive

(A) >870 models screened and passed QC for 8-point dose-response curves at 3x dilution from 10 μ M for each compound.

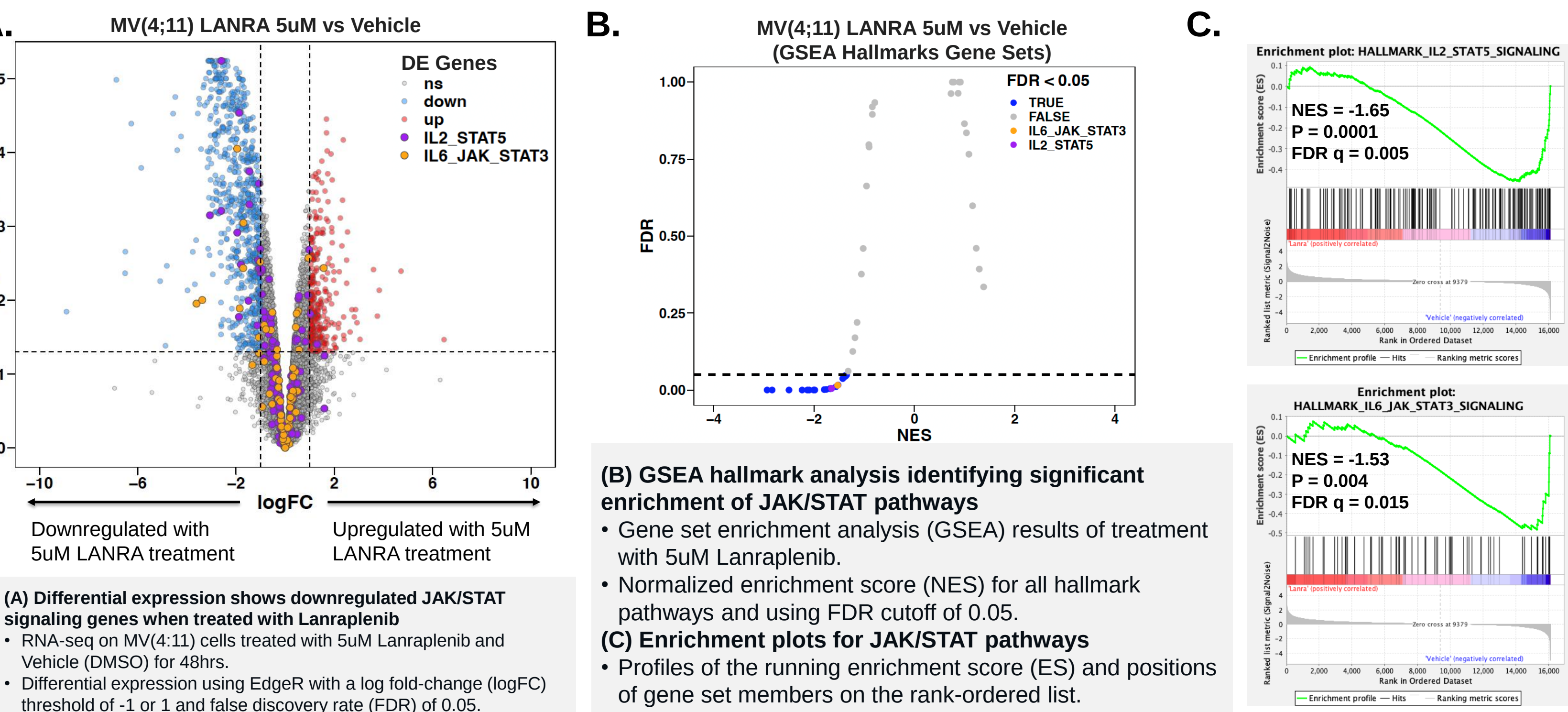
(B) Among the combos tested, Ruxolitinib and Lanraplenib showed enriched activity in convergent lineages.

(C) Across all models and lineages, the combination showed increased anti-proliferative effect at both Lanraplenib dose levels. Two anchor doses of Lanraplenib at 2 μ M and 10 μ M were profiled against a 7-point titration of Ruxolitinib.

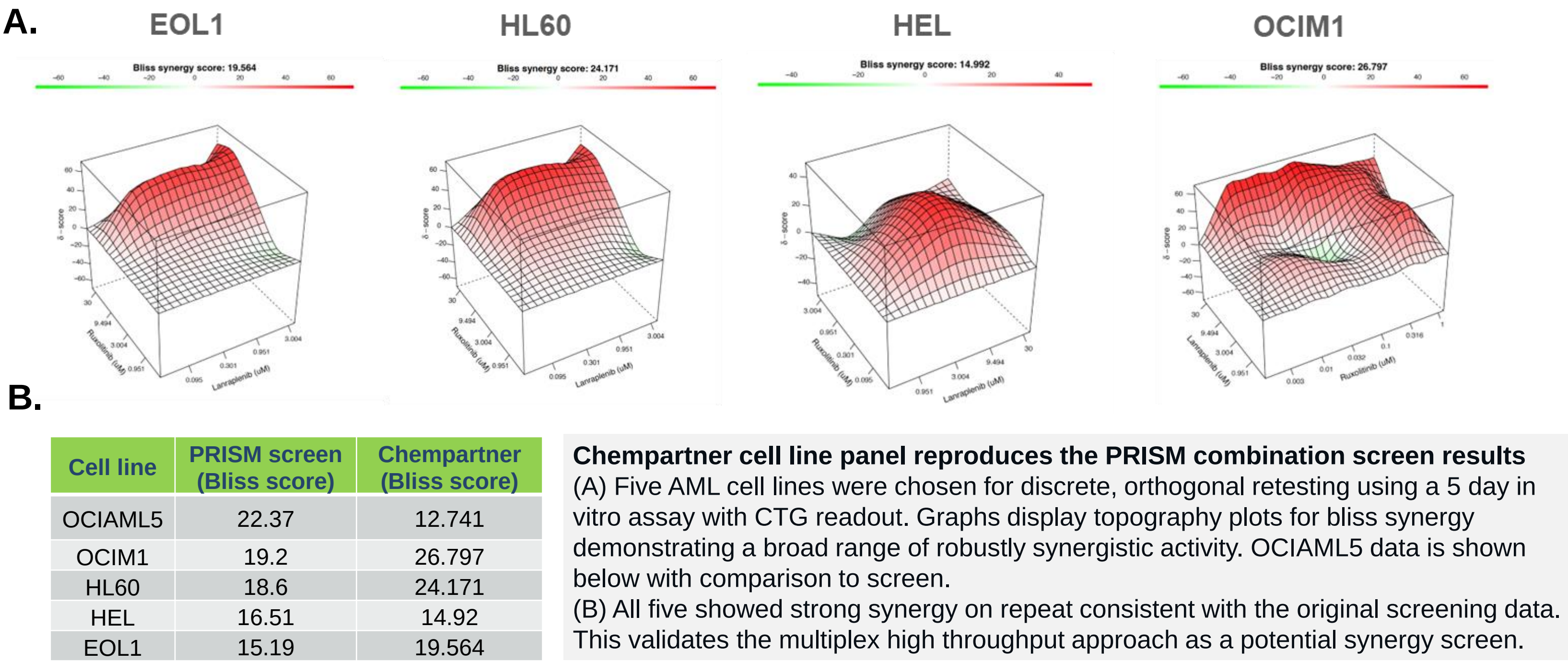
PRISM combination effect is most strongly observed in hematological malignancies



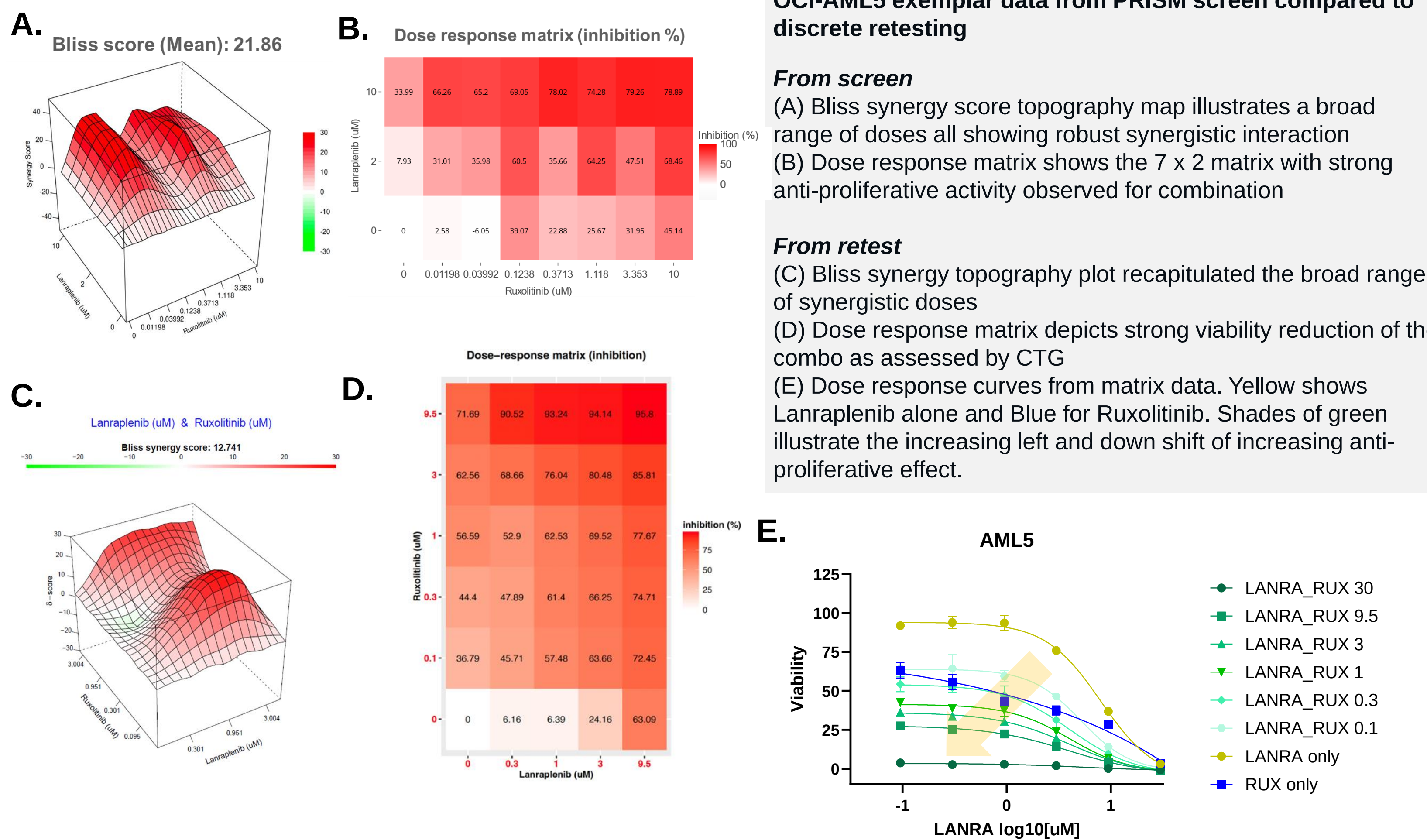
JAK/STAT signaling pathways are downregulated in response to Lanraplenib



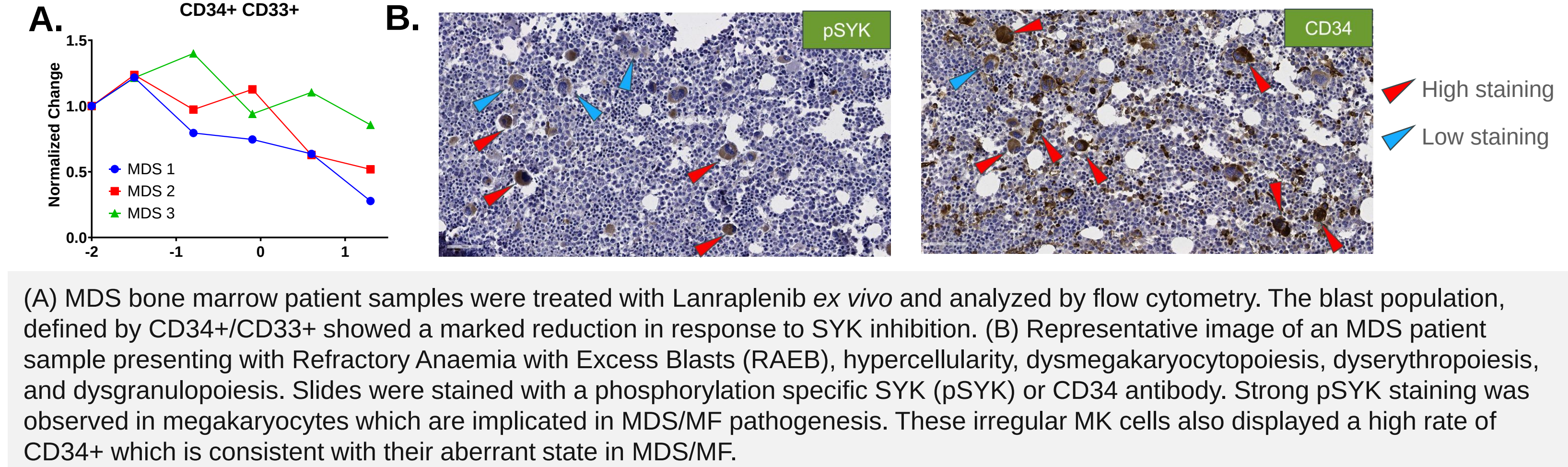
Ruxolitinib synergizes with Lanraplenib in targeted AML cell line panel



Orthogonal retesting validates screening data and approach



Myeloid dysplastic Syndrome (MDS) patient samples are responsive to Lanraplenib



Conclusions

These studies demonstrate the utility of PRISM as a platform to rapidly identify rational combination agents. Importantly, Lanraplenib is effective in combination with ruxolitinib in AML and other hematological malignancy preclinical models. This finding is consistent with the observation that SYK can regulate STAT signaling and cooperate with other RTKs like FLT3. Given the central role of SYK in regulation of oncogenic and inflammatory signaling, SYK inhibition with Lanraplenib in combination with ruxolitinib may be a promising strategy for patients with myeloid malignancies.

References

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3. Sprissler C et al. Blood Cancer Journal (2014).
4. Yu C et al. Nature Biotechnology (2016).